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Journal of International Money and Finance

journal homepage: www.elsevier.com/locate/jimf



Out-of-sample bond risk premium predictions: A global common factor[☆]



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ARTICLE INFO

Article history:

Available online 20 November 2014

JEL classifications:

G1

E4

F3

Keywords:

Bond risk premia

Economic value

Global common factor

Return predictability

Out-of-sample forecasts

ABSTRACT

This paper investigates the out-of-sample predictability of international bond risk premia. We endogenously construct a global common [Cochrane and Piazzesi \(2005\)](#) factor. We find that the global factor strongly predicts international bond risk premia and delivers economically significant gains relative to the historical average. The forecasting power of the global factor is above and beyond the predictive power contained in country-specific factors. As predicted by economic theories, bond return forecasts appear countercyclical. We also find that the global factor is related to international economic activity.

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1. Introduction

The issue of bond return predictability is of great interest to academics and practitioners. Accordingly, a vast amount of research has been devoted to predicting excess bond returns. Traditional literature focuses on the predictive ability of forward rates for future excess bond returns. Because forward rates represent the rate one can lock in now for a commitment to buy a one period bond in the

[☆] We thank Kees Koedijk (the editor), an anonymous referee, Guofu Zhou, Yangru Wu, Victoria Long, Claire Santos, Cynthia Deng, Michael Bell, Rahman Shihidur, Choy Keen Meng, Dennis Alba, and seminar participants at the World Banking and Finance Symposium 2013, the Singapore Economic Conference 2013, Aarhus University, Central University of Finance and Economics, Nanyang Technological University, and Shanghai University of Finance and Economics. Financial support from the Natural Science Foundation of China (Grant No. 71473281) is acknowledged.

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future, it is natural to hypothesize that they incorporate useful information for predicting future bond returns. In the US market, many studies (e.g., Fama and Bliss, 1987; Campbell and Shiller, 1991; Cochrane and Piazzesi, 2005) reveal that forward rates are able to generate accurate forecasts of excess bond returns. In international markets, several studies (e.g., Dahlquist and Hasseltoft, 2013; Kessler and Scherer, 2009) find that forward rates strongly predict international excess bond returns.

We contribute to the existing literature in three ways: First, we endogenously construct a global common Cochrane and Piazzesi (2005, CP hereafter) factor, which is a linear combination of international forward rates. Second and foremost, we evaluate the predictive power of the global common CP factor in an out-of-sample forecasting exercise. Though a few studies (e.g., Ilmanen, 1995; Dahlquist and Hasseltoft, 2013) reveal that some global common factors forecast international bond risk premia in-sample, little is known about the out-of-sample predictability of excess bond returns in international markets. Since a model's in-sample predictive ability tends to correlate poorly with its out-of-sample forecasting performance (e.g., Thornton and Valente, 2012; Inoue and Kilian, 2004, 2006), it is important to evaluate the out-of-sample predictability of international excess bond returns. Indeed, the out-of-sample forecasting analysis can sharpen our understanding on the global driving force of excess bond returns. Third, because statistical significance does not mechanically lead to economic significance, we assess the economic value of the global common CP factor in asset allocation.

We analyze the predictable variation in bond risk premia from January 1980 to December 2011 in the four largest and most liquid markets: the United States, the United Kingdom, Germany, and Japan. We endogenously construct the global CP factor by regressing the average of annual international excess returns on the country-specific CP factors. Our empirical analysis proceeds in three steps.

First, we investigate the statistical significance of the global CP factor in bond return predictions. The out-of-sample forecasting analysis shows that the global CP factor consistently delivers significant out-of-sample gains relative to the historical average.¹ More importantly, we find that the forecasting power of the global factor is above and beyond the predictive power contained in the country-specific CP factors. Indeed, the global factor strongly predicts international bond risk premia with the out-of-sample R^2 up to 42.8%. The average out-of-sample R^2 is 19.4%. These results are robust in subsamples. These results are also robust in international data, which are not used to construct the global CP factor.

Second, we move beyond comparing forecasting performance from a statistical perspective and evaluate the economic gain of the global CP factor from an asset allocation perspective. Specifically, we quantify how much a risk-averse investor is willing to pay to switch from a dynamic portfolio strategy based on the no-predictability benchmark to a model that uses the information content of the global factor. We find that the information content of the global CP factor does generate systematic economic value to investors.

Third, we investigate the nature of the predictive ability of the global factor. Rational asset pricing theories (e.g., Campbell and Cochrane, 1999; Wachter, 2006; Buraschi and Jiltsov, 2007) usually imply that return predictability can result from exposure to time-varying aggregate risk or time-varying risk aversion. Intuitively, heightened risk aversion during economic recessions demands a higher risk premium. It thus generates return predictability. To provide some insight on the economic interpretation of the global factor, we link the global factor to international economic activity. In line with Kojen et al. (2012) and Dahlquist and Hasseltoft (2013), we find that the global factor predicts international industrial production growth, suggesting that the global factor contains important information about macroeconomic fundamentals. The empirical analysis also suggests that bond risk premia predicted by the global CP factor appear countercyclical in all four markets. This finding is consistent with economic theories suggesting that investors must be compensated with risks associated with economic recessions.

On the theoretical side, how to understand the forecasting power of the global factor across markets? Rizova (2010) shows that a two-country, Lucas-tree framework with gradual information diffusion (e.g., Hong and Stein, 1999; Hong et al., 2007; Rapach et al., 2013) can cause asset returns in one country to predict asset returns in a trading-partner country. In the context of international bond risk premium predictability, information about global economic fundamentals diffuses simultaneously

¹ It is consistent with the expectations hypothesis of interest rates, which states that the premium on long-term bonds over short-term bonds is constant over time.

to each other, so forward rates in one market have predictive power for bond risk premia in other markets. As such, [Rizova \(2010\)](#) provides an economic interpretation for the predictive ability of the global factor since the global factor synthesizes information about the behavior of international forward rates.

This paper is closely related to [Ilmanen \(1995\)](#), who finds that some global instruments can predict international bond risk premia better than country-specific local factors. Our paper is also related to [Dahlquist and Hasseltoft \(2013\)](#), who construct the global factor as a GDP-weighted average of the country-specific CP factors and investigate the in-sample forecasting performance of the GDP-weighted factor. We differ from [Dahlquist and Hasseltoft \(2013\)](#) by endogenously constructing the global factor and by examining the out-of-sample forecasting performance and the economic value of the global factor. Our economic analysis framework taken here is similar to the framework in [Goh et al. \(2013\)](#). However, we focus on the predictive ability of forward rates in international markets. In contrast, [Goh et al. \(2013\)](#) focus on the forecasting power of technical indicators in the US market.²

The remainder of the article is organized as follows: Section 2 reviews the literature. Section 3 discusses the way of constructing the global CP factor. Section 4 conducts the out-of-sample forecasting exercise and reports the empirical results. Section 5 examines the link between the global factor and international economic activity. The section also investigates links between bond return forecasts and real economic activity. Section 6 concludes.

2. Literature review

In last two decades, economists have devoted substantial attention to predicting bond risk premia. To date, three groups of factors have been identified as predictors of bond risk premia. They are respectively term structure factors, macro factors, and financial variables.

The intuition for using term structure factors to predict bond risk premia is straightforward: investors' beliefs about future bond prices determine what investors are willing to pay for bonds. This indicates that current term structure contains all information relevant to predicting bond risk premia. Empirically, this view is supported by many studies (e.g., [Keim and Stambaugh, 1986](#); [Fama and Bliss, 1987](#); [Stambaugh, 1988](#); [Fama and French, 1989](#); [Ilmanen, 1995](#); [Cochrane and Piazzesi, 2005, 2008](#)), which reveal that various term structure factors such as the yield spread, forward rates, and real interest rate can predict bond risk premia.

The classic [Merton \(1971\)](#) model provides the theoretical foundation for the state of economy or the changing investment opportunity set as the source of time-varying risk premia. Insofar as macro variables affect future consumption and investment opportunities, they are important state variables for predicting excess bond returns. In this spirit, a number of studies (e.g., [Ilmanen, 1995](#); [Ludvigson and Ng, 2009](#); [Cooper and Priestley, 2009](#); [Joslin et al., 2014](#); [Zhou and Zhu, 2014](#)) show that macro factors have important forecasting power for future bond risk premia.

Bonds and interest rate derivatives are actively traded in financial markets. Naturally, financial variables related to trading activity of these financial instruments should contain valuable information about bond risk premia because the price and trading volume of these instruments are sensitive to underlying interest rates. Motivated by this argument, a number of empirical studies³ investigate the predictive power of financial factors and identify several predictors of bond risk premia such as interest rate option prices, futures market interest, the maturity of debt issues, and the variance risk premium.

3. The CP factor

This section constructs the global and country-specific CP factors. Specifically, Section 3.1 describes the approach for constructing the country-specific CP factors. Section 3.2 discusses data issues and

² While the motivation for using technical indicators to predict bond risk premia is behavioral, the motivation for using forward rates is rational.

³ For example, [Baker et al. \(2003\)](#), [Almeida et al. \(2011\)](#), [Goh et al. \(2013\)](#), [Hong and Yogo \(2012\)](#), [Muller et al. \(2011\)](#), [Wright and Zhou \(2009\)](#).

presents summary statistics. Section 3.3 introduces the endogenous way of constructing the global CP factor. The section also reports the empirical results of estimating the global and country-specific CP factors.

3.1. The country-specific CP factor

This section describes the approach for constructing country-specific CP factors. We follow [Cochrane and Piazzesi \(2005\)](#) for the notation of excess bond returns and yields. Let $p_t^{(n)}$ denote the log price of n -year discount bond at time t , the log yield is $y_t^{(n)} \equiv -\frac{1}{n}p_t^{(n)}$. The log one-year forward rate between periods $t + n - 1$ and $t + n$ is $f_t^{(n)} \equiv p_t^{(n-1)} - p_t^{(n)}$. The excess return of an n -year bond is computed as the log-holding period return from buying an n -year bond at time t and selling it after one year less the yield on a one-year bond at time t .

$$rx_{t+1}^{(n)} \equiv p_{t+1}^{(n-1)} - p_t^{(n)} - y_t^{(1)}. \quad (1)$$

The average excess log return across the maturity spectrum is defined as.

$$\bar{rx}_{t+1} \equiv \frac{1}{4} \sum_{n=2}^5 rx_{t+1}^{(n)}.$$

According to finance theories, investors' beliefs about future bond prices are impounded into current prices. As such, the time- t term structure should contain substantial information about future changes in excess bond returns. Since forward rates are the rate at which investors commit to buy a one-period bond at a future date, it is naturally to hypothesize that forward rates can predict excess bond returns. In this spirit, [Cochrane and Piazzesi \(2005\)](#) extend the [Fama and Bliss \(1987\)](#) approach and run regressions of excess returns on all forward rates:

$$rx_{t+1}^{(n)} = \beta_0^{(n)} + \beta_1^{(n)} y_t^{(1)} + \beta_2^{(n)} f_t^{(2)} + \dots + \beta_5^{(n)} f_t^{(5)} + \varepsilon_{t+1}^{(n)}. \quad (2)$$

Drawing on the fact that the same function of forward rates predicts holding period returns at all maturities, [Cochrane and Piazzesi \(2005\)](#) construct a tent-shaped linear combination of forward rates, namely the CP factor, to parsimoniously predict future excess bond returns. Specifically, the CP factor is constructed by regressing the average excess return across maturities at each time t on the one-year yield and four forward rates $\mathbf{f}_t \equiv [y_t^{(1)} \ f_t^{(2)} \ f_t^{(3)} \ f_t^{(4)} \ f_t^{(5)}]'$:

$$\bar{rx}_{t+1} = \gamma_0 + \gamma' \mathbf{f}_t + \bar{\varepsilon}_{t+1}, \quad (3)$$

where $\gamma = [\gamma_1 \ \gamma_2 \ \gamma_3 \ \gamma_4 \ \gamma_5]'$. The CP factor is the fitted value of regression (3).

3.2. Data

The data set is made up of monthly observations of the end-of-month 1-year, 2-year, 3-year, 4-year, and 5-year zero-coupon interest rates for the United States (US), the United Kingdom (UK), Japan (JP), and Germany (GM), the four largest bond markets in the world. The data set extends from January 1980 to December 2011. The standard point of data availability determines the starting period of the data.⁴ The US data are obtained from the Fama-Bliss discount bond file in the Center for Research in Securities Prices (CRSP). We collect the UK data from the Bank of England. The Japanese data are obtained from

⁴ The Japanese data are available after January 1980.

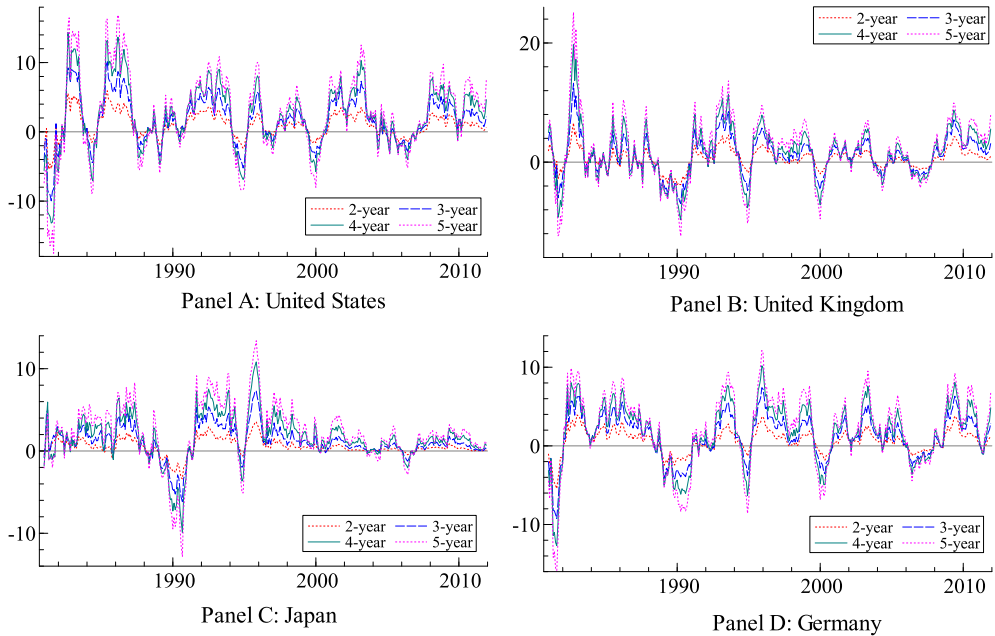


Fig. 1. Time series of international excess bond returns. The sample period is 1980:01 to 2011:12. The excess returns plotted in these panels include the risk premia on the two-, three-, four-, and five-year bonds over the one-year bond. Excess bond returns are expressed in percentage points per annum.

the Ministry of Finance Japan. The German observations are retrieved from the Bundesbank. All interest rates are denominated in local currency.⁵

Excess bond returns are calculated from zero-coupon bond yields as described in equation (1). Fig. 1 plots international excess returns. We also report the summary statistics of the excess bond returns in Table 1. It appears that international bond risk premia show some similar patterns. Bond risk premia are all statistically significant at the 5% confidence level. Though the average of bond risk premia is small, bond risk premia are very volatile, as indicated by Fig. 1. On average, bond risk premia increase along the maturity spectrum in all bond markets. Table 1 also reports the persistence of bond risk premia. The autocorrelation coefficients of order 1 indicate that bond risk premia are generally persistent. However, the autocorrelation coefficients of order 12 are often negative or insignificant, suggesting that bond risk premia may show the mean-reverting behavior. This is indicative that bond risk premia may have a predictable component.

Since bonds are traded internationally, bond risk premia across markets are likely to be highly correlated. As such, there is likely a global common factor driving international bond risk premia. Before embarking on the construction of a global common factor, it is interesting to examine the correlation between international bond risk premia. Table 2 reports the correlation coefficients of the average of bond risk premia in each market. The correlation coefficients range between 0.31 and 0.73,

⁵ This implicitly means that international investors hedge currency risk in portfolio management (e.g., Ang and Bekaert, 2002). To illustrate the point, let y_t , y_t^* , s_t respectively denote the one-period domestic log interest rate, foreign log interest rate, and log exchange rate. The log excess return denominated in domestic currency for a domestic investor who invests in the foreign bond market is $rx_{t+1}^* = s_{t+1} - s_t + Y_{t+1}^* - y_t$, where $Y_{t+1}^* = p_{t+1}^{(n-1)} - p_t^n$ is the gross return denominated in foreign currency for holding a n -period foreign bond for one-period. If investors hedge currency risk using currency futures, no-arbitrage pricing implies that $s_{t+1} - s_t = y_t - y_t^*$. Finally, we have $rx_{t+1}^* = p_{t+1}^{(n-1)} - p_t^n - y_t^*$, which is equivalent to equation (1).

Table 1
Summary statistics of international excess bond returns.

Maturity	Central moments		Autocorrelations	
	Mean	Std dev	Lag 1	Lag 12
US				
$rx_t^{(2)}$	0.97***	1.74***	0.927***	0.003**
$rx_t^{(3)}$	1.72***	3.30***	0.929***	−0.068
$rx_t^{(4)}$	2.36***	4.65***	0.929***	−0.109
$rx_t^{(5)}$	2.69***	5.76***	0.914***	−0.155*
UK				
$rx_t^{(2)}$	0.58***	1.62***	0.925***	0.034***
$rx_t^{(3)}$	1.15***	3.01***	0.919***	−0.102***
$rx_t^{(4)}$	1.63***	4.27***	0.915***	−0.185***
$rx_t^{(5)}$	2.06***	5.37***	0.911***	−0.221***
Japan				
$rx_t^{(2)}$	0.53***	0.98***	0.923***	0.264**
$rx_t^{(3)}$	1.11***	1.88***	0.919***	0.227*
$rx_t^{(4)}$	1.66***	2.77***	0.915***	0.199
$rx_t^{(5)}$	2.20***	3.56***	0.922***	0.176
Germany				
$rx_t^{(2)}$	0.63***	1.46***	0.951***	0.073**
$rx_t^{(3)}$	1.25***	2.70***	0.946***	0.030**
$rx_t^{(4)}$	1.74***	3.75***	0.943***	−0.005*
$rx_t^{(5)}$	2.12***	4.67***	0.940***	−0.032

The table summarizes the descriptive statistics for international excess bond returns. The data sample ranges from January 1980 to December 2011. The excess bond returns are expressed in percentage points per annum. *, **, and *** denote statistical significance at the 10%, 5%, or 1% confidence level, and statistical significance is evaluated using the Newey–West (1987) autocorrelation and heteroscedasticity-consistent standard errors.

indicating that bond risk premia are significantly correlated. This may indicate the presence of a global common factor.

3.3. The global CP factor

The Treasury bond market is affected by relatively few factors. There is no default risk or cash flow uncertainty. In addition, almost all foreign exchange risk can be hedged. The simplicity of Treasury

Table 2
Correlation coefficients between average excess bond returns.

	$\overline{rx}_{t+1}^{US}$	$\overline{rx}_{t+1}^{UK}$	$\overline{rx}_{t+1}^{JP}$	$\overline{rx}_{t+1}^{GM}$
$\overline{rx}_{t+1}^{US}$	1.00	—	—	—
$\overline{rx}_{t+1}^{UK}$	0.57	1.00	—	—
$\overline{rx}_{t+1}^{JP}$	0.31	0.48	1.00	—
$\overline{rx}_{t+1}^{GM}$	0.66	0.73	0.53	1.00

The table reports the pairwise correlation coefficients between the average excess bond returns on government bonds for US, UK, Japan, and Germany. The sample period is 1980:01 to 2011:12. The average excess return in each market is calculated as

$$\overline{rx}_{t+1} \equiv \frac{1}{4} \sum_{n=2}^5 rx_{t+1}^{(n)}.$$

Table 3

Correlation coefficients between the global and country-specific factors.

	CP_t^G	CP_t^{US}	CP_t^{UK}	CP_t^{JP}	CP_t^{GM}
CP_t^G	1.00	–	–	–	–
CP_t^{US}	0.67	1.00	–	–	–
CP_t^{UK}	0.67	0.40	1.00	–	–
CP_t^{JP}	0.42	0.02	0.21	1.00	–
CP_{t+1}^{GM}	0.60	0.28	0.47	0.31	1.00

The table reports the pairwise correlation coefficients between the global and country-specific CP factors. The sample period is 1980:01 to 2011:12.

bonds facilitates the construction of a global common CP factor. Specifically, to construct the global Cochrane–Piazzesi factor, we define the international average excess return across markets as⁶

$$\hat{r}x_{t+1} \equiv \frac{1}{4} (\bar{r}x_{t+1}^{US} + \bar{r}x_{t+1}^{UK} + \bar{r}x_{t+1}^{JP} + \bar{r}x_{t+1}^{GM}).$$

The CP factor is constructed by regressing international average excess returns at each time t on the four country-specific CP factors (CP_t^C):

$$\hat{r}x_{t+1} = \gamma_0 + \gamma_{US} CP_{t+1}^{US} + \gamma_{UK} CP_{t+1}^{UK} + \gamma_{JP} CP_{t+1}^{JP} + \gamma_{GM} CP_{t+1}^{GM} + \hat{\varepsilon}_{t+1}. \quad (4)$$

The global CP factor is the fitted value of the regression

$$CP_{t+1}^G = \gamma_0 + \gamma_{US} CP_{t+1}^{US} + \gamma_{UK} CP_{t+1}^{UK} + \gamma_{JP} CP_{t+1}^{JP} + \gamma_{GM} CP_{t+1}^{GM}.$$

We apply our method to empirically construct the global and country-specific CP factors. Table 3 reports the correlation coefficients between the global and country-specific factors. These factors are found to be positively correlated. Among the country-specific factors, the correlation coefficients range from 0.02 (between the US and JP factors) to 0.47 (between the UK and GM factors). In contrast, the global factor is more significantly correlated with the country-specific factors with the JP factor being least correlated with the global factor.

In the construction of the global factor, a natural concern caused by the correlated country-specific factors is multicollinearity. To shed light on this issue, we conduct the eigensystem analysis of the four country-specific CP factors. The condition number of the regressors is found to be 10.8, suggesting that there is no serious problem with multicollinearity.

4. Out-of-sample predictability

Welch and Goyal (2008) argue that out-of-sample tests seem to be a more relevant standard for assessing genuine return predictability in real time. In the spirit of Welch and Goyal (2008), we conduct the out-of-sample exercise to investigate the forecasting ability of the global CP factor.⁷

4.1. Predictive regressions

Following Cochrane and Piazzesi (2005), we begin with a predictive regression model for excess bond returns, which can be expressed as.

⁶ We also construct the global CP factor by regressing the GDP-weighted average of international excess returns on the four country-specific factors. We obtain similar results.

⁷ An earlier version of this paper presents the in-sample results of predicting international bond risk premia using the global factor. The in-sample results are similar to those reported in Dahlquist and Hasseltoft (2013).

$$r\mathbf{x}_{c,t+1}^{(n)} = \alpha_{c,n} + \beta'_{c,n}F_t + \varepsilon_{c,t+1}^{(n)}, \quad \text{for } c = \text{US, UK, JP or GM.} \quad (5)$$

where F_t denotes predictive variables whose predictive ability is of interest, and $\varepsilon_{c,t+1}^{(n)}$ is a disturbance term. In this paper, we consider three combinations of predictive variables: (i) F_t only includes a country-specific CP factor; (ii) F_t only includes the global CP factor; (iii) F_t includes both the global and country-specific factors. In so doing, we attempt to provide insight on the absolute and additional predictive ability of the global factor. To avoid look-ahead bias and the use of future data, we generate out-of-sample forecasts of excess bond returns using recursive predictive regressions, with all predictive factors recursively constructed and parameters estimated just using information available up to the month of forecast formation. More specifically, we estimate and forecast recursively, using data from 1980:01 to the time that the forecast is made, beginning in 1992:01 and extending through 2011:12.

When $\beta_{c,n} = 0$, excess bond returns are equal to a constant $\alpha_{c,n}$, which is exactly the historical average of excess returns $\bar{r}\mathbf{x}_{c,t}^{(n)}$. Following [Cochrane and Piazzesi \(2005\)](#) and [Thornton and Valente \(2012\)](#), we use this historical average of excess bond returns to serve as a natural benchmark forecasting model. Indeed, the historical average is consistent with the expectations hypothesis of the term structure of interest rates, which is typically used as a benchmark against which other empirical forecasting models are compared.

4.2. Evaluation

To evaluate the out-of-sample forecasting performance of various models, we use the out-of-sample R^2 statistic, R_{OS}^2 , suggested by [Campbell and Thompson \(2008\)](#) to compare forecasts $\bar{r}\mathbf{x}_{c,t+1}^{(n)}$ and $\hat{r}\mathbf{x}_{c,t+1}^{(n)}$, where $\bar{r}\mathbf{x}_{c,t+1}^{(n)}$ is the historical average, and $\hat{r}\mathbf{x}_{c,t+1}^{(n)}$ is an individual forecast based on predictive regression (5). The R_{OS}^2 statistic is akin to the familiar in-sample R^2 statistic and is given by.

$$R_{OS}^2 = 1 - \frac{\sum_{k=t}^T \left(r\mathbf{x}_{c,t+1}^{(n)} - \hat{r}\mathbf{x}_{c,t+1}^{(n)} \right)^2}{\sum_{k=t}^T \left(r\mathbf{x}_{c,t+1}^{(n)} - \bar{r}\mathbf{x}_{c,t+1}^{(n)} \right)^2}.$$

The R_{OS}^2 statistic measures the reduction in mean square prediction error (MSPE) for the predictive regression model relative to the historical average forecast. When $R_{OS}^2 > 0$, the $\hat{r}\mathbf{x}_{c,t+1}^{(n)}$ forecast outperforms the historical average forecast according to the MSPE metric.

The *** statistic is a point estimate of forecast accuracy. To test whether two competing models have statistically significant difference in forecasting accuracy, [Clark and West \(2007\)](#) develop an MSPE-adjusted statistic to test the null hypothesis of $R_{OS}^2 \geq 0$ against the alternative of $R_{OS}^2 < 0$. The Clark-West statistic is conveniently computed by first defining.

$$f_{t+1} = \left(r\mathbf{x}_{c,t+1}^{(n)} - \bar{r}\mathbf{x}_{c,t+1}^{(n)} \right)^2 - \left[\left(r\mathbf{x}_{c,t+1}^{(n)} - \hat{r}\mathbf{x}_{c,t+1}^{(n)} \right)^2 - \left(\bar{r}\mathbf{x}_{c,t+1}^{(n)} - \hat{r}\mathbf{x}_{c,t+1}^{(n)} \right)^2 \right]. \quad (6)$$

By regressing $\{f_{s+1}\}_{s=t}^{T-1}$ on a constant and computing the t -statistic corresponding to the constant, a p -value for a one-sided (upper tail) test is obtained with the standard normal distribution.

R_{OS}^2 statistics are typically large for bond risk premium forecasts, but a relatively large R_{OS}^2 does not mechanically imply significant economic value for an investor (e.g., [Thornton and Valente, 2012](#); [Zhou and Zhu, 2014](#)). It is therefore important to assess the economic value of the global factor. To shed light on this issue, following [Kandel and Stambaugh \(1996\)](#), [Campbell and Thompson \(2008\)](#) and [Welch and Goyal \(2008\)](#), realized utility gain is used to measure economic significance. Based on the certainty equivalence, [Kandel and Stambaugh \(1996\)](#) and [Campbell and Thompson \(2008\)](#) show that, for a

mean-variance investor with a power utility function, the utility of a trading strategy is equal to expected portfolio return less $(\gamma/2)$ times portfolio variance, where γ can be interpreted as the coefficient of relative risk aversion. To measure utility gain, the first step is to measure the average utility of the trading strategy based on the historical average. According to the mean-variance framework, the investor will decide at the end of period t to allocate the following share of his or her portfolio to risky long-term bond in period $t + 1$:

$$\bar{w}_t = \left(\frac{1}{\gamma}\right) \left(\frac{\bar{r}x_{c,t+1}^{(n)}}{\hat{\sigma}_{t+1}^2}\right), \quad (7)$$

where $\hat{\sigma}_t^2$ is the ten-year rolling-window estimate of the variance of excess returns.⁸ Over the out-of-sample period, the investor realizes an average utility level of

$$\bar{v} = \bar{\mu} - \left(\frac{1}{2}\right) \gamma \bar{\sigma}^2, \quad (8)$$

where $\bar{\mu}$ and $\bar{\sigma}^2$ are the sample mean and variance, respectively, over the out-of-sample period for the return on the benchmark portfolio formed using forecasts of excess bond returns based on the historical average.

The second step is to calculate the average utility of the trading strategy based on predictive regression (5). Similarly, the investor will choose a long-term bond share of.

$$\hat{w}_t = \left(\frac{1}{\gamma}\right) \left(\frac{\hat{r}x_{c,t+1}^{(n)}}{\hat{\sigma}_{t+1}^2}\right), \quad (9)$$

and realizes an average utility level of

$$\hat{v} = \hat{\mu} - \left(\frac{1}{2}\right) \gamma \hat{\sigma}^2, \quad (10)$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ are the sample mean and variance, respectively, over the out-of-sample period for the return on the portfolio formed using the predictive regression (5). The economic value of predictive variables is the difference between $\bar{v}$ and $\hat{v}$.

4.3. Empirical results

Table 4 presents the Campbell and Thompson (2008) R_{OS}^2 statistics for out-of-sample excess return forecasts over the 1992:01–2011:12 evaluation period. Panel A reports the results for the excess bond return forecasts based on the country-specific CP factors relative to the historical average benchmark forecast. The results suggest that the country-specific factors consistently predict bond risk premia, with the R_{OS}^2 statistics ranging between 5.3% and 34.2%. We assess the statistical significance of the R_{OS}^2 statistics using the Clark and West (2007) MSPE-adjusted test. The testing results show that the R_{OS}^2 statistics are consistently significant at the 5% level. In addition, we conduct an economic test by evaluating the profitability of an asset-allocation strategy that uses the information content of the country-specific CP factors. The performance measures are calculated for the out-of-sample period by assuming $\gamma = 5$, in line with the value used in previous studies (e.g., Thornton and Valente, 2012).⁹ Our

⁸ $w_{0,t}$ is restricted to be positive and less than 150% at maximum so that extreme investments are prevented, as in Campbell and Thompson (2008) and Welch and Goyal (2008).

⁹ The results are qualitatively similar for other reasonable γ values.

Table 4

Out-of-sample forecasts of international bond risk premia.

	US		UK		Japan		Germany	
	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)
Panel A: Country-specific factor								
$rx_t^{(2)}$	6.7***	0.74	34.2***	1.08	22.0***	0.36	14.6***	1.84
$rx_t^{(3)}$	11.7***	1.49	29.5***	1.49	21.1***	0.37	14.9***	1.83
$rx_t^{(4)}$	7.4***	1.88	17.2***	1.58	23.2***	0.38	18.6***	1.65
$rx_t^{(5)}$	5.7***	2.25	5.3***	1.67	18.7***	0.31	18.7***	1.53
Panel B: Global factor								
$rx_t^{(2)}$	4.5***	1.11	37.3***	1.23	−3.0**	0.22	17.5***	1.46
$rx_t^{(3)}$	11.7***	1.57	37.5***	1.81	−0.8**	0.18	31.2***	1.75
$rx_t^{(4)}$	12.6***	1.82	33.1***	2.18	6.5***	0.21	37.3***	1.68
$rx_t^{(5)}$	13.3***	2.21	27.4***	2.29	12.3***	0.27	42.8***	1.80
Panel C: Country-specific and global factors								
$rx_t^{(2)}$	4.6***	1.21	[39.5]***	1.14	29.5***	0.41	[29.6]***	1.52
$rx_t^{(3)}$	11.9***	1.78	[36.6]***	1.77	27.3***	0.42	[38.3]***	1.84
$rx_t^{(4)}$	10.4***	2.08	[27.1]***	2.10	{32.1}***	0.43	[42.2]***	1.95
$rx_t^{(5)}$	{10.7}***	2.46	[17.8]***	2.18	[28.4]***	0.39	[43.5]***	2.06

This table reports the out-of-sample R^2 statistics for log excess bond returns on the n -year long-term Treasury bond over the 1992:01–2011:12 forecast evaluation period. Utility gain (Δ) is the portfolio management fee (in annualized percentage return) that an investor would be willing to pay to have access to the forecasting model. Panels A, B, and C respectively present the results for the forecasting model with the country-specific factor, the global factor, and both factors. Statistical significance for the out-of-sample R^2 is based on the p -value for the Clark and West (2007) out-of-sample MSPE-adjusted statistic. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. "{ }" and "[]" mean that the two-predictor model significantly outperforms the local-factor only model at the 10% and 1% significance levels.

empirical analysis reveals that the country-specific CP factors generate systematic economic value, ranging between 2.25% and 0.31%.¹⁰

Panel B reports the out-of-sample results for the bond return forecasts based on the global CP factor relative to the historical average benchmark forecast. The global factor generally provides greater support for out-of-sample predictability, as 11 out of the 16 excess return forecasts produce a higher R^2_{OS} statistic than the country-specific CP factors do. The global CP factor consistently delivers economic value in excess return forecasts, suggesting that the global CP factor is an economically meaningful predictor for international bond risk premia.

Indeed, our findings are consistent with those reported in some earlier studies. Ilmanen (1995) and Zhou and Zhu (2014) reveal that some global instruments predict international excess bond returns better than local instruments. Harvey (1991) also finds that global common factors outperform local factors in the prediction of international excess stock returns. These findings may reflect the integrated nature of international bond markets. In an integrated market, global factors are the driving force of international bond risk premia. As such, global factors are more likely to predict excess bond returns in international markets.

One may concern that the information content of the global factor is subsumed by country-specific CP factors. To provide insight on this important issue, we use two predictors in predictive regression (5): the global and country-specific factors. If the two-predictor model generates higher R^2_{OS} statistics and economic value, it means that the global factor has significant forecasting power for international excess returns, above and beyond the predictive power contained in country-specific factors.

¹⁰ At first sight, our results are inconsistent with those reported in Thornton and Valente (2012). However, it seems that there are structural breaks in the Thornton and Valente sample. When we use the Thornton and Valente sample to check the robustness of our forecasting procedure, the forecasting analysis reveals that economic value is negligible. However, the R^2_{OS} statistics are still significantly positive.

Table 5

Out-of-sample forecasts of bond risk premia: subsample analysis.

	US		UK		Japan		Germany	
	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)
Panel A: Country-specific factor								
$rx_t^{(2)}$	-7.2***	0.22	14.5***	0.20	31.3***	0.10	16.9***	1.04
$rx_t^{(3)}$	4.92***	0.25	22.3***	0.51	26.6***	0.02	26.5***	1.37
$rx_t^{(4)}$	4.8***	0.32	20.5***	1.14	16.5***	-0.08	32.9***	1.65
$rx_t^{(5)}$	8.5***	0.45	13.3***	1.63	23.2***	-0.14	36.8***	1.78
Panel B: Global factor								
$rx_t^{(2)}$	22.5***	1.23	26.9***	1.61	2.2	0.09	18.0***	0.98
$rx_t^{(3)}$	25.8***	1.21	39.6***	2.01	3.7*	0.07	20.6***	1.34
$rx_t^{(4)}$	25.4***	1.17	44.3***	2.30	5.3**	0.14	20.6***	1.51
$rx_t^{(5)}$	25.5***	1.16	43.6***	2.59	6.7**	0.17	19.0***	1.67
Panel C: country-specific and global factors								
$rx_t^{(2)}$	{2.3}***	0.72	[25.9]***	1.63	{33.4}***	0.16	[20.2]***	1.57
$rx_t^{(3)}$	{6.8}***	0.78	[38.3]***	1.95	[32.2]***	0.21	26.3***	2.18
$rx_t^{(4)}$	{7.9}***	0.92	[41.2]***	2.29	[20.8]***	0.24	{30.2}***	2.44
$rx_t^{(5)}$	[13.5]***	1.02	[37.9]***	2.57	[12.9]***	0.25	[31.5]***	2.63

This table reports the out-of-sample R^2 statistics for log excess bond returns on the n -year long-term Treasury bond over the 2008:01–2011:12 forecast evaluation period. Utility gain (Δ) is the portfolio management fee (in annualized percentage return) that an investor would be willing to pay to have access to the forecasting model. Panels A, B, and C respectively present the results for the forecasting model with the country-specific factor, the global factor, and both factors. Statistical significance for the out-of-sample R^2 is based on the p -value for the Clark and West (2007) out-of-sample MSPE-adjusted statistic. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. "{ }" and "[]" mean that the two-predictor model significantly outperforms the local-factor only model at the 10% and 5% significance levels.

Panel C presents the out-of-sample forecasting results for the two-predictor model. It shows that the two-predictor model consistently produces large R_{OS}^2 statistics relative to the historical average benchmark for excess returns on Treasury bonds. In 15 out of the 16 excess return forecasts, the two-predictor model generates higher R_{OS}^2 statistics than the " CP_t^c -only" model does. These results suggest that the global factor has additional forecasting power for excess bond returns. Furthermore, we assess the statistical significance of the higher R_{OS}^2 statistics of the two-predictor model relative to the CP_t^c -only model using the Clark and West (2007) MSPE-adjusted statistic. We find that, in 11 out of 15 excess return forecasts, the better out-of-sample forecasting performance of the two-predictor model is statistically significant, suggesting that the global factor is a powerful predictor of international bond risk premia. The MSPE-adjusted statistics also suggest that the information content of international forward rates is very important for predicting UK and GM bond risk premia. In contrast, it is less important for predicting US and JP bond risk premia.

Panel C also reports utility gains of the two-predictor model. We find that the two-predictor model consistently delivers positive economic value. Compared with utility gains generated by the CP_t^c -only model, we find that, in 15 out of the 16 excess return forecasts, the two-predictor model delivers higher utility gains. Taken together with the R_{OS}^2 statistics, it is safe to say that the global factor contains information about future excess bond returns, above and beyond the forecasting power contained in country-specific factors.

Our results indicate that excess bond returns are much more predictable than stock risk premia in terms of out-of-sample R_{OS}^2 . While the out-of-sample R_{OS}^2 statistic for predicting excess stock returns is typically less than 5% (e.g., Welch and Goyal, 2008; Campbell and Thompson, 2008; Cooper and Priestley, 2009), our analysis reveals that the average out-of-sample R_{OS}^2 statistic for predicting bond risk premia is 19.4% in international bond markets. These results are largely consistent with other studies (e.g., Ludvigson and Ng, 2009; Goh, Jlang, Tu, and Zhou, 2013; Zhou and Zhu, 2014) that have looked at the out-of-sample predictability of bond returns. A natural question is why bond risk premia

are much more predictable than excess stock returns. One potential explanation is that bond risk premia are highly persistent, as indicated in Table 1. In contrast, excess stock returns nearly follow random walks. Though the R^2_{OS} statistic in the bond market is much greater than that in the stock market, it is important to note that the economic value of excess bond return forecasts is only comparable to that of excess stock return forecasts (e.g., Campbell and Thompson, 2008; Rapach et al., 2013). This is consistent with intermarket efficiency, which suggests that the economic value of forecasting is likely to be similar due to cross market arbitrage.

4.4. Subsample analysis

Welch and Goyal (2008) find that the predictability of excess returns seems to be unstable or even spurious. In his webpage, John Cochrane suggests that the predictive power of the CP factor seems to be weak during the recent 2007–2009 financial crisis. To provide insight on the robustness of the forecasting performance of the global factor over different sample periods, we repeat our out-of-sample forecasting analysis using the financial crisis subsample. Specifically, we use 1990:01–2007:12 as the initial in-sample period when forming out-of-sample forecasts of excess bond returns for the period 2008:01–2011:12. We conduct the out-of-sample forecasting exercise recursively to avoid look-ahead bias. Specifically, forecasts are generated using parameters that are estimated using information available only at the time the forecast is made. Producing forecasts in this manner simulates the situation of an investor in real time. We assess the statistical significance of R^2_{OS} using the Clark and West (2007) MSPE-adjusted statistic.

Table 5 reports the empirical results. Panel A reveals that the country-specific CP factors perform well over the out-of-sample period: the R^2_{OS} statistics are consistently positive and statistically significant at the 5% level. At the same time, the country-specific factors usually deliver positive economic value. Notably, while the Japanese CP factor performs remarkably well in term of R^2_{OS} statistic, the economic value of utilizing information from the Japanese CP factor is very small or even negative.

Panel B compares the relative forecasting performance of the global CP factor and the historical average. According to the R^2_{OS} statistics, the forecasts utilizing information from the global factor are superior to the historical average, with the R^2_{OS} statistics ranging between 6.9 percent and 49.7 percent. The Clark-West CSPE-adjusted test indicates that these positive R^2_{OS} statistics are consistently significant. In addition, the economic analysis reveals that the global factor consistently generates positive economic value, suggesting that the global factor contains economically important information about future bond risk premia.

When utilizing information contained in the global and country-specific factors together, the predictive model performs very well. In 13 out of 16 excess returns, the two-predictor forecasts significantly outperform the return forecasts based solely on the country-specific factors, suggesting that the two-predictor model generally beats the CP^c_t -only model. In addition, the two-predictor model consistently generates higher economic value. When the competitive model is the CP^c_t -only model, the average utility gain of the global factor is respectively 1.49% and 1.98% in the US and UK bond market. These results indicate that the global CP factor plays an important role in forecasting international bond risk premia.

4.5. Predicting bond risk premia in other markets

The global CP factor is extracted from forward rates in four major bond markets: the United States, the United Kingdom, Japan, and Germany. A natural question is whether the global CP factor can predict bond risk premia in other bond markets. Obtaining evidence on the forecasting power of the global factor in other markets is important because it provides insight on how common is the global common factor. Toward this end, we repeat the out-of-sample forecasting analysis using government bond yields for Switzerland and Sweden.¹¹ Yields for Switzerland are derived from forward rates. The

¹¹ We deliberately omit Euro-zone countries because their term structures have been highly correlated with the Germany term structure, though they diverged for a while during the 2007–2009 financial crisis.

data start in January 1988 and end in December 2011. Yields for Sweden are obtained from the Riksbank and Datastream. The data start in January 1994 and end in December 2011.

We employ a recursive forecasting scheme. Specifically, forecasts are generated using parameters that are estimated using information available only at the time the forecast is made. The out-of-sample period is 2000:01–2011:12 for Switzerland and 2004:01–2011:12 for Sweden. Economic value is computed for the out-of-sample period by assuming $\gamma = 5$.

Table 6 reports the forecasting results. Panel A presents the out-of-sample forecasting results for the CP_t^c -only model. The R_{OS}^2 statistics reveal that the country-specific CP factors significantly predict excess bond returns, with the R_{OS}^2 statistics ranging between 21.9% and 41.9% for Switzerland and between 39.4% and 54.5% for Sweden. The economic analysis further suggests that the country-specific factors generally deliver positive economic value, the only exception is the excess return on the Swiss 2-year government bond.

Panel B reports the forecasting results using the global CP factor as the predictive variable. The results indicate that the global CP factor strongly predicts excess bond returns in Switzerland and Sweden, with the R_{OS}^2 statistics ranging between 21.5% and 52.2%, though the construction of the global factor uses no information contained in forward rates from Switzerland and Sweden. The R_{OS}^2 statistics are consistently significant at the 5% level. Moreover, the global factor generally produces positive economic value. Taken together, these results point to a leading role for US, UK, JP, and GM in international bond markets.

Panel C of Table 6 reports the out-of-sample forecasting results for the two-predictor model. It is evident that the two-predictor model consistently delivers positive and highly significant R_{OS}^2 statistics, which are generally higher than those delivered by the country-specific factors. The MSPE-adjusted statistics reveal that the greater predictability of the two-predictor model is usually significant at the 10% or 1% significance level. The economic analysis further shows that the two-predictor model

Table 6
Out-of-sample forecasts of bond risk premia: Additional results.

	Switzerland		Sweden	
	R^2 (%)	Δ (%)	R^2 (%)	Δ (%)
Panel A: Country-specific factor				
$rx_t^{(2)}$	41.9***	−0.59	53.2***	0.97
$rx_t^{(3)}$	34.7***	0.39	54.5***	1.57
$rx_t^{(4)}$	28.1***	0.73	49.1***	1.75
$rx_t^{(5)}$	21.9***	0.87	39.4***	1.89
Panel B: Global factor				
$rx_t^{(2)}$	48.7***	−0.58	21.5***	0.42
$rx_t^{(3)}$	52.2***	0.62	26.2***	1.27
$rx_t^{(4)}$	51.5***	1.09	25.8***	1.49
$rx_t^{(5)}$	47.9***	1.53	29.9***	1.81
Panel C: two factors				
$rx_t^{(2)}$	[52.5]***	−0.55	54.3***	1.05
$rx_t^{(3)}$	[51.3]***	0.65	55.9***	1.72
$rx_t^{(4)}$	[47.2]***	1.18	51.3***	2.04
$rx_t^{(5)}$	[41.6]***	1.36	[44.1]***	2.43

This table reports the out-of-sample R^2 statistics for log excess bond returns on the n -year long-term Treasury bond. Utility gain (Δ) is the portfolio management fee (in annualized percentage return) that an investor would be willing to pay to have access to the forecasting model. Panels A, B, and C respectively present the results for the forecasting model with the country-specific factor, the global factor, and both factors. Statistical significance for the out-of-sample R^2 is based on the p -value for the Clark and West (2007) out-of-sample MSPE-adjusted statistic. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. “[] ” and “[] ” mean that the two-predictor model significantly outperforms the local-factor only model at the 10% and 1% significance levels.

generally generates positive economic value, which is consistently greater than the economic value delivered by the CP_t^c -only model. Overall, these results seem to suggest that the global factor contains additional information for predicting excess bond returns in Switzerland and Sweden.

4.6. Estimation risk and economic value

Without knowledge of the true parameter values, investors face parameter uncertainty (alias *estimation risk*) when constructing their optimal portfolio. In this section, we check the robustness of our results by incorporating estimation risk into portfolio decisions. Toward this end, it is natural to take a Bayesian approach (e.g., Barberis, 2000; Della Corte et al., 2008; Pástor and Stambaugh, 2009). In so doing, the uncertainty about the predictive regression parameters is summarized by the posterior distribution of the parameters given the data. Rather than constructing the distribution of future excess bond returns conditional on fixed parameter estimates, the Bayesian method taken here integrates over the uncertainty in the parameters captured by the posterior distribution.

To construct the posterior distribution of the parameters, we use a conventional uninformative prior. Appendix A provides the detail of the Bayesian method for estimating the predictive model. We calculate the economic value of various predictive models using the posterior distributions. Table 7 reports the economic gains for the global predictive model, the CP_t^c -only model, and the two-factor model. It seems that investors usually benefit from recognizing estimation risk. Importantly, the economic gains of the two-factor model are generally higher than the economic gains of the CP_t^c -only model, suggesting that the global factor is important for making portfolio decisions.

4.7. International diversification with currency hedging

Having demonstrated that the global CP factor delivers significant benefits to investors who invest in a single bond market, we now address the question of the economic benefit of the global factor in international asset allocation for investors who make investments in international bond markets. We evaluate the economic benefit of the global factor in international asset allocation by quantifying how much a risk-averse investor is willing to pay to switch from a dynamic portfolio strategy based on the predictions of the CP_t^c -only model to a model that uses the global CP factor. Since currency hedging may affect return predictability (e.g., Kilian and Taylor, 2003; Wright, 2008; Campbell et al., 2003), our international diversification analysis takes currency hedging into consideration.

If we use currency futures to hedge currency risk, it can be shown (see footnote ⁵) that the excess hedged foreign bond returns are given by Equation (1), where $p_{t+1}^{(n)}$ and $y_t^{(1)}$ are denominated in local currency. Now suppose a US investor is making investment in international bond markets (US, UK, Japan, and Germany). Our empirical analysis shows that the utility difference between the global CP factor investing strategy and the country-specific CP factor strategy is 1.85%,¹² suggesting that the global CP factor delivers economic gains in international portfolio diversification.

5. Economic links

The predictive power of the global factor raises a natural question: What is the information source of the global factor? In this section, we investigate the information source of the global factor. Specifically, we focus on two aspects: links between the global factor and international economic activity, and links between bond return forecasts and the real economy.

Examining links among the global factor, return forecasts, and economic activity is motivated by economic theories suggesting that rational, utility-maximizing investors must be compensated for bearing macroeconomic risks. For example, in the Campbell and Cochrane (1999) habit model framework, Buraschi and Jiltsov (2007) and Wachter (2006) link term structure dynamics to economic

¹² The economic gains of the global factor in international asset allocation are quantitatively similar for an investor in UK, Japan, or Germany.

Table 7

The economic value of the global factor in Internatoinal diversification.

	Local CP factor				Global CP factor				Two-factor model			
	$rx_t^{(2)}$	$rx_t^{(3)}$	$rx_t^{(4)}$	$rx_t^{(5)}$	$rx_t^{(2)}$	$rx_t^{(3)}$	$rx_t^{(4)}$	$rx_t^{(5)}$	$rx_t^{(2)}$	$rx_t^{(3)}$	$rx_t^{(4)}$	$rx_t^{(5)}$
US	0.86	1.63	2.03	2.18	1.17	1.65	1.90	2.25	1.33	1.86	2.14	2.48
UK	1.04	1.51	1.66	1.72	1.32	1.79	2.21	2.26	1.18	1.85	2.07	2.20
JP	0.43	0.47	0.45	0.33	0.25	0.20	0.18	0.25	0.57	0.60	0.58	0.55
GM	1.77	1.80	1.69	1.49	1.52	1.71	1.75	1.79	1.63	1.88	1.87	1.97

This table respectively reports the utility gain (Δ) for the forecasting model with the country-specific factor, the global factor, and both factors over the 1992:01–2011:12 forecast evaluation period.

fundamentals and show that bond risk premia should vary with macroeconomic shocks. In a recursive utility framework, Piazzesi and Schneider (2007) suggest that investors may require a premium to hold nominal bonds due to the fear of stagflation. Empirically, it has been well-documented that the information content of the term structure can predict economic activity. For example, Estrella and Hardouvelis (1991) reveal that the yield spread can predict economic growth. Ang et al. (2006) demonstrate that short-term rates strongly predict GDP growth. Recently, Kojen et al. (2012) and Dahlquist and Hasseltoft (2013) demonstrate that the CP factor can predict economic activity.

5.1. The global factor and economic activity

In this section, we use the global CP factor to predict international economic activity as measured by annual industrial production growth (IP). The investigation of economic links sheds light on the economic source of the predictive power of the global factor. The regression is

$$IP_{t+1}^c = \alpha_c + \beta_c CP_t^G + \varepsilon_{c,t+1}, \quad \text{for } c = \text{US, UK, JP or GM.} \quad (11)$$

Table 8 reports the results for predicting international economic activity. Panel A presents the regression slope coefficients, heteroscedasticity and serial correlation robust t -statistics, and R^2 s. We find that the global CP factor is generally significant at the 5% level. The positive coefficients suggest that an increase in global risk premia is followed by improving economic conditions. According to economic theories, investors require higher risk premia during economic recessions. While economic recessions lead an increase in risk premia, a rise in risk premia tends to lead improvement in future economic growth (e.g., Ang et al., 2006).

5.2. Bond return forecasts and the real economy

The findings presented so far reveal that the global factor strongly predicts international bond risk premia, but they do not tell us whether bond return forecasts are countercyclical, as predicted by rational asset pricing theories. To address this issue, we investigate links between bond return forecasts and economic activity.

Table 8

The global factor and international economic activity.

Country	β_1	t -value	R^2
Panel A: Forecasts of IP			
US	0.95	(6.69)	0.20
UK	0.58	(4.74)	0.11
Japan	0.30	(1.26)	0.01
Germany	0.35	(1.98)	0.03

The table presents the results of predicting international economic activity using the global factor. The sample period is 1980:01–2011:12. Point estimates are reported with Newey and West (1987) t -statistics, accounting for conditional heteroscedasticity and serial correlation up to 18 lags, in parentheses.

Table 9

Correlation coefficients between return forecasts and detrended IP.

	$rx_{t+1}^{(2)}$	$rx_{t+1}^{(3)}$	$rx_{t+1}^{(4)}$	$rx_{t+1}^{(5)}$
US	−0.299	−0.298	−0.294	−0.294
UK	−0.231	−0.240	−0.244	−0.245
JP	−0.445	−0.435	−0.422	−0.419
GM	−0.394	−0.397	−0.395	−0.392

The table reports the pairwise correlation coefficients between the out-of-sample bond return forecasts and the detrended industrial production for US, UK, Japan, and Germany. The sample period is 1992:01 to 2011:12.

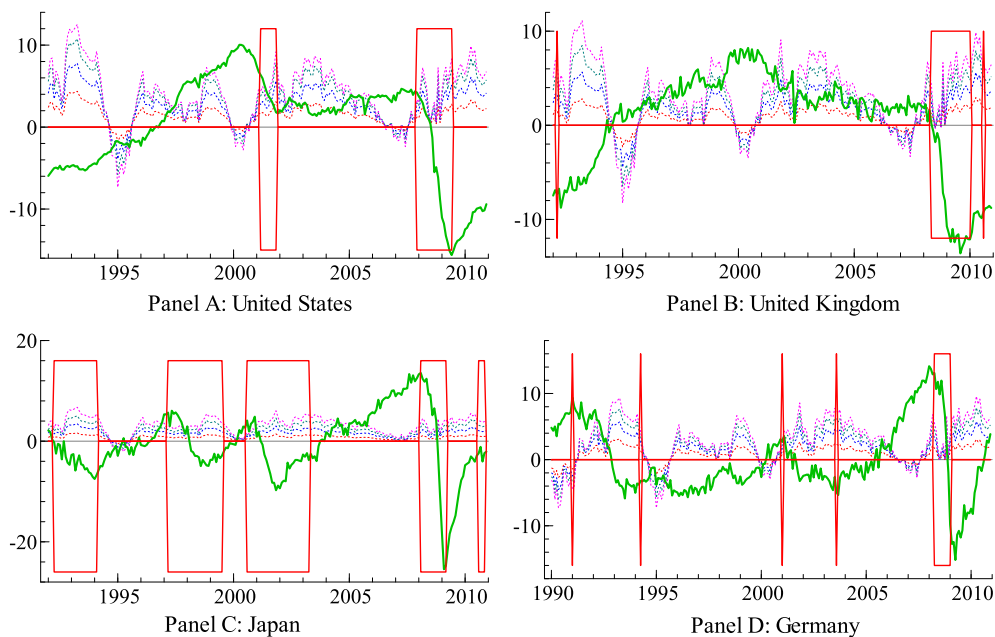


Fig. 2. Bond return forecasts and detrended industrial production. The solid lines are detrended industrial production. The dotted lines (from the lowest to the highest line, with occasional cross-overs) are the annual excess returns of the two-, three-, four-, and five-year bonds over the one-year bond. The bars indicate economic recessions.

Table 9 reports the pairwise correlation coefficients between the out-of-sample bond return forecasts predicted by the global factor and detrended industrial production. The out-of-sample period is the 1992:01–2011:12 evaluation period. All correlation coefficients are negative. On average, the correlation coefficients for JP and GM are higher than those for US and UK. These results clearly indicate that bond return forecasts are countercyclical.

To further shed light on links between the bond return forecasts and economic activity, Fig. 2 plots the bond return forecasts and IP_t over time, along with shaded bars indicating business-cycle recessions.¹³ The figure shows significant patterns in return forecasts around economic recessions. Specifically, there are distinct upward spikes in bond return forecasts at or shortly after the troughs in the US, UK, and GM bond markets. The JP bond market shows extraordinary patterns largely because Japan went through a ‘Great Recession’ starting in 1990. Overall, our analysis indicates that bond return forecasts are countercyclical.

¹³ We use NBER-dated business recessions for the United States. For all other three markets, the business cycle is dated by the Economic Cycle Research Institute (ECRI).

6. Conclusions

We endogenously construct a global CP factor and examine the predictability of international government bond risk premia. We find that the global CP factor has economically and statistically significant forecasting power for excess bond returns. Importantly, the predictive ability of the global factor is above and beyond the ability contained in the country-specific factors. As predicted by economic theories, we find that bond return forecasts appear countercyclical.

We also investigate economic gains accruing to an investor who exploits the predictability of bond excess returns. In particular, we quantify how much a risk-averse investor is willing to pay to switch from a dynamic portfolio strategy based on the historical average model to a portfolio strategy based on the global factor. Our economic analysis suggests that predicting excess bond returns using the global factor is economically meaningful.

Identifying a global common factor that predicts international bond risk premia has three direct benefits. First, it gives us a better understanding on the global driving force of the time-varying bond risk premia. Second, it sharpens our understanding on the comovement of international bond markets. Third, it may indicate investment opportunities in international bond markets. For future research, we will conduct the analysis using repo rates (e.g., [Zhu, 2014](#)).

Appendix A

We can write predictive regression (5) as

$$Y = X\beta + E, \quad (\text{A-1})$$

where $Y = \{r_{c,t+1}^{(n)}\}_{t=1}^T$ is a $T \times 1$ vector of excess returns, $X = \{1, F_t'\}_{t=0}^{T-1}$ is a $T \times k$ matrix of predictors, $\beta = [\alpha_{c,n}, \beta'_{c,n}]'$ is a $k \times 1$ vector of parameters, and $E = \{\varepsilon_{c,t+1}^{(n)}\}_{t=1}^T$ is a $T \times 1$ vector of error terms with $E \sim N(0, \sigma^2)$. The parameter set we need to estimate is $\theta = [\beta', \sigma^2]'$.

Bayesian estimation requires three elements: the data, prior distributions for the model's parameters, and posterior distributions. Let $D = [Y, X]$ denote the data, $p(D|\theta)$ denote the likelihood function, and $\pi(\theta)$ denote the prior distribution for the parameter set θ . Following Bayes rule, the posterior distribution is proportional to the product of the likelihood function and the prior distribution on the parameters.

$$\pi(\theta) \propto p(D|\theta)\pi(\theta).$$

Bayesian estimation is accomplished by analyzing the joint posterior distribution of the parameter set θ .

A.1. Prior distributions

Prior distributions summarize economic agent's prior beliefs about θ . In our empirical applications, we independently draw parameters β and σ^2 . This implies that the parameters can be factored as $\pi(\theta) = \pi(\beta)\pi(\sigma^2)$. It is possible to model prior beliefs by using any reasonable distribution form. Our strategy is to use fairly standard conjugate prior distributions for β and σ^2 . These conjugate prior distributions lead to well-defined posterior distributions (e.g., [Zellner, 1971](#)). Specifically, we choose

$$\begin{aligned} \pi(\beta) &\sim N(\beta_0, \Sigma_0^{-1}), \\ \pi(\sigma^2) &\sim iG\left(\frac{r_0}{2}, \frac{s_0}{2}\right), \end{aligned}$$

where N denotes a k -variate normal distribution, and iG denotes the inverted Gamma distribution. As for the prior parameters, we set $\beta_0 = 0$, $\Sigma_0^{-1} = I_k$, $r_0 = 2$, and $s_0 = 1$. By setting $\beta_0 = 0$, our prior belief is

that excess bond returns are unpredictable. This implies that our prior belief is uninformative about future excess bond returns.

A.2. Posterior distributions

Given the prior distributions and predictive regression (A-1), we have the following posterior distribution for β :

$$\pi(\beta|\sigma^2, D) \sim N(\beta_*, \Sigma_*), \quad (\text{A-2})$$

where $\Sigma_* = \Sigma_0^{-1} + \frac{1}{\sigma^2} X'X$, and $\beta_* = \Sigma_* \left(\frac{1}{\sigma^2} X'Y + \Sigma_0^{-1} \beta_0 \right)$. The posterior distribution for σ^2 is

$$\pi(\sigma^2|\beta, D) \sim iG\left(\frac{r_*}{2}, \frac{s_*}{2}\right), \quad (\text{A-3})$$

where $r_* = T + r_0$, and $s_* = (Y - X\beta)'(Y - X\beta)$.

A.3. Bayesian estimation (Gibbs sampling)

We use the Gibbs sampling method to draw parameters from the posterior distributions. In our application, the Gibbs sampler proceeds as follows:

- (1) Initialize β and σ^2 .
- (2) Draw β from the posterior distribution (A-2).
- (3) Draw σ^2 from the posterior distribution (A-3).
- (4) Go to step 2 and repeat.

After discarding the first 20,000 iterations, we collect the subsequent 100,000 draws to calculate the posterior mean and variance of the parameters.

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